GOVERNING EQUATIONS AND OPENFOAM RUN ANALYSIS

Comparative Verification of Dimensional and Kinematic Formulations

1. Stokes Flow · 2. Newtonian Navier–Stokes · 3. Herschel–Bulkley Rheology

Author: Danish Durrany Zulfahmy
Project: CFD Analysis of Cuttings Transport in an Eccentric Horizontal Wellbore ($e = 0.623$, 60 RPM)
Computational Platform: OpenFOAM v2406 (Ubuntu WSL2) · ParaView 6.2.0
Academic Session: 2026/2027

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1 Introduction & Problem Formulation

Accurate prediction of solid–liquid cuttings transport and bed erosion in an eccentric wellbore ($R_{\text{wb}} = 0.05$ m, $R_p = 0.025$ m, $e = 0.623$) requires a stepwise numerical verification workflow. Before activating full Eulerian–Eulerian multiphase coupling with solid particulate interactions, three fundamental continuum models were developed, simulated, and post-processed:

1. **Method 1: Stokes Flow (`stokesCavity`)** — Creeping viscous flow regime ($\text{Re} \ll 1$) evaluated to confirm finite volume spatial discretization and numerical stability ($\text{Co} = 0.20$).
2. **Method 2: Incompressible Newtonian Navier–Stokes (`eccentricNS_clean`, `icoFoam`)** — Introduction of non-linear convective acceleration $(\mathbf{U} \cdot \nabla)\mathbf{U}$ and rotating wall kinematic boundary conditions ($\Omega = 60$ RPM $= 6.2832$ rad/s).
3. **Method 3: Incompressible Herschel–Bulkley Fluid (`eccentric_HB`, `pimpleFoam`)** — Implementation of yield-stress shear-thinning mud rheology to resolve both the high-shear annular lubricated boundary layer and the unyielded plug core in the wider gap.

2 Dimensional Derivations vs. OpenFOAM Kinematic Formulation

2.1 Fundamental Momentum Conservation (Dimensional Form)

In continuum mechanics, the momentum conservation equation for an incompressible fluid is written in its classical **dimensional form** (units of force per unit volume, N/m^3) as:

$$\rho \left(\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} \right) = -\nabla P + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g} \tag{1}$$

where:

- ρ is the fluid density (kg/m^3),
- $\mathbf{U}$ is the instantaneous velocity vector field (m/s),
- P is the static thermodynamic pressure ($\text{Pa} = \text{N/m}^2$),
- $\boldsymbol{\tau}$ is the viscous stress tensor (Pa),
- $\mathbf{g}$ is the gravitational acceleration vector (m/s^2).

For a standard Newtonian fluid with dynamic viscosity μ (Pa·s), the stress divergence simplifies to $\nabla \cdot \boldsymbol{\tau} = \mu \nabla^2 \mathbf{U}$, yielding:

$$\rho \left(\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} \right) = -\nabla P + \mu \nabla^2 \mathbf{U} + \rho \mathbf{g} \tag{2}$$

2.2 Kinematic Reduction in Incompressible OpenFOAM

For isothermal single-phase flow where fluid density ρ remains strictly constant throughout the domain, OpenFOAM's incompressible solvers (`icoFoam`, `pimpleFoam`) divide Equation (1) through by constant ρ:

$$\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla \left(\frac{P}{\rho} \right) + \nabla \cdot \left(\frac{\boldsymbol{\tau}}{\rho} \right) + \mathbf{g} \tag{3}$$

OpenFOAM defines internal solver variables directly on a kinematic basis:

$$p = \frac{P}{\rho} \quad \left[\frac{\mathrm{m}^2}{\mathrm{s}^2} \right], \qquad \nu = \frac{\mu}{\rho} \quad \left[\frac{\mathrm{m}^2}{\mathrm{s}} \right] \tag{4}$$

Furthermore, in single-phase wall-bounded horizontal cross-sections of constant density, gravitational acceleration produces a conservative hydrostatic gradient ($\nabla(\rho \mathbf{g} \cdot \mathbf{x})$) that generates zero baroclinic vorticity ($\nabla \rho \times \mathbf{g} = \mathbf{0}$) and does not distort the shear field. As a result, OpenFOAM solves the streamlined kinematic equation:

$$\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla p + \nu \nabla^2 \mathbf{U} \tag{5}$$

2.3 Multiphase Generalization: Cuttings Transport Formulation

In the subsequent multiphase phase of this research (`twoPhaseEulerFoam`), density ρ and gravity $\mathbf{g}$ cannot be divided out. The density differential between the solid rock cuttings ($\rho_s \approx$ 2600–2760 kg/m^3) and the carrier drilling mud ($\rho_l \approx$ 1000–1200 kg/m^3) produces the net settling and buoyancy body forces:

$$\mathbf{F}_{\text{net}} = \frac{\pi}{6} d_p^3 (\rho_s - \rho_l)\mathbf{g} \tag{6}$$

The full phase-coupled momentum conservation equations are therefore resolved in complete dimensional form:

$$\frac{\partial (\alpha_l \rho_l \mathbf{U}_l)}{\partial t} + \nabla \cdot (\alpha_l \rho_l \mathbf{U}_l \mathbf{U}_l) = -\alpha_l \nabla P + \nabla \cdot (\alpha_l \boldsymbol{\tau}_l) + \alpha_l \rho_l \mathbf{g} + \mathbf{M}_{sl} \tag{7}$$

$$\frac{\partial (\alpha_s \rho_s \mathbf{U}_s)}{\partial t} + \nabla \cdot (\alpha_s \rho_s \mathbf{U}_s \mathbf{U}_s) = -\alpha_s \nabla P + \nabla \cdot (\alpha_s \boldsymbol{\tau}_s) + \alpha_s \rho_s \mathbf{g} - \mathbf{M}_{sl} \tag{8}$$

where α_l and α_s are the respective liquid and solid phase volume fractions ($\alpha_l + \alpha_s = 1$), and $\mathbf{M}_{sl}$ represents the interphase momentum transfer (drag, virtual mass, and lift).

Model / Case	Dimensional Form ($\mathbf{N/m^3}$)	OpenFOAM Kinematic Form ($\mathbf{m/s^2}$)	Rheological Closure / Mechanism
1. Stokes Flow (`stokesCavity`)	$-\nabla P + \mu\nabla^2\mathbf{U} = \mathbf{0}$ Inertia $\rho(\mathbf{U}\cdot\nabla)\mathbf{U} \approx \mathbf{0}$	$-\nabla p + \nu\nabla^2\mathbf{U} = \mathbf{0}$	Constant scalar: $\nu = 1.0 \times 10^{-2}$ m^2/s ($\mu = \rho\nu$); symmetric stagnation/suction pressure peaks ($p \in [-24, +25]$).
2. Newtonian NS (`eccentricNS_clean`)	$\rho\left(\frac{\partial\mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U}\right) = -\nabla P + \mu\nabla^2\mathbf{U}$	$\frac{\partial\mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U} = -\nabla p + \nu\nabla^2\mathbf{U}$	Constant scalar: $\nu = 1.0 \times 10^{-4}$ m^2/s; broad Couette velocity decay; hydrodynamic pressure wedge ($p \in [-7.8 \times 10^{-4}, +6.4 \times 10^{-3}]$).
3. Herschel–Bulkley (`eccentric_HB`)	$\rho\left(\frac{\partial\mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U}\right) = -\nabla P + \nabla\cdot\boldsymbol{\tau}$ $\boldsymbol{\tau} = \mu_{\text{eff}}(\dot\gamma)\left[\nabla\mathbf{U} + (\nabla\mathbf{U})^T\right]$	$\frac{\partial\mathbf{U}}{\partial t} + (\mathbf{U}\cdot\nabla)\mathbf{U} = -\nabla p + \nabla\cdot\left(\nu_{\text{eff}}\left[\nabla\mathbf{U} + (\nabla\mathbf{U})^T\right]\right)$	Dynamic scalar field: $\nu_{\text{eff}} = \min\left(\nu_0, \frac{\tau_0}{\dot\gamma} + k\dot\gamma^{n-1}\right)$, where $\tau_0 = \tau_y/\rho$, $k = K/\rho$. Localized shear liquefaction with wide-gap unyielded plug core.

Table 1: Systematic comparison of governing equations in classical dimensional form versus OpenFOAM kinematic implementation.

3 OpenFOAM Dictionaries & Numerical Setup

3.1 Herschel–Bulkley Transport Properties (`constant/transportProperties`)

```
transportModel  HerschelBulkley;
nu              [0 2 -1 0 0 0 0] 1e-4;

HerschelBulkleyCoeffs
{
    nu0         [0 2 -1 0 0 0 0] 1e-2;   // Yielded plug limiting viscosity (m2/s)
    nuInf       [0 2 -1 0 0 0 0] 1e-5;   // Infinite-shear limiting viscosity (m2/s)
    k           [0 2 -1 0 0 0 0] 4e-4;   // Kinematic consistency index K/rho
    n           [0 0 0 0 0 0 0]  0.65;   // Flow behavior index (shear-thinning n < 1)
    tau0        [0 2 -2 0 0 0 0] 4e-3;   // Kinematic yield stress tau_y/rho (m2/s2)
}
```

Listing 1: OpenFOAM Herschel-Bulkley kinematic transport configuration

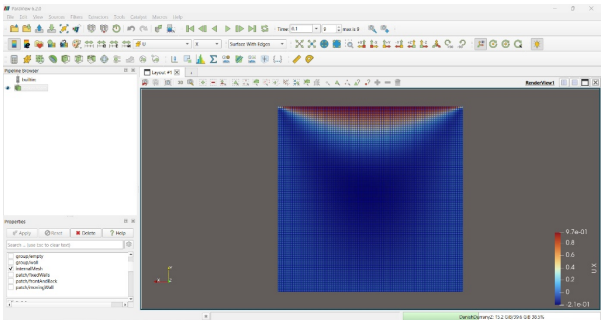
3.2 Rotating Drillpipe Boundary Condition (`0/U`)

```
boundaryField
{
    outerWall
    {
        type            noSlip;
    }
    innerWall
    {
        type            rotatingWallVelocity;
        origin          (0 -0.015575 0); // Downward offset for e = 0.623
        axis            (0 0 1);
        omega           6.28318;         // 60 RPM counter-clockwise
    }
    frontAndBack
    {
        type            empty;
    }
}
```

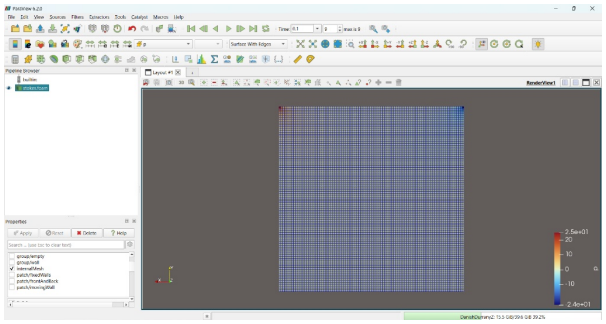
Listing 2: Rotating wall condition for eccentric inner cylinder

4 ParaView Post-Processing & Comparative Analysis

4.1 Method 1: Stokes Cavity Flow Verification



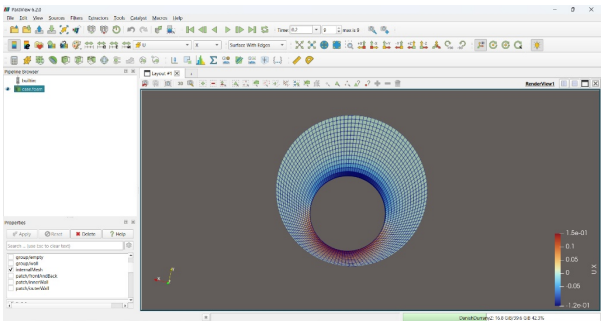
(a) Horizontal Velocity $U_x \in [-0.21, +0.97]$ m/s



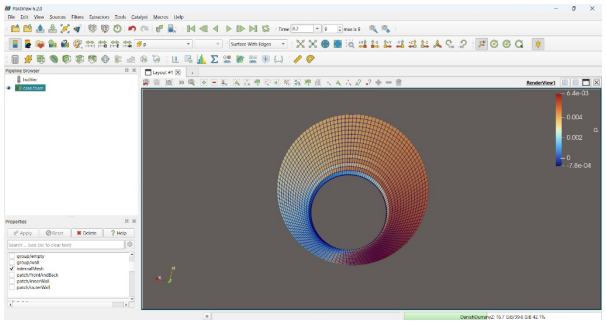
(b) Kinematic Pressure $p \in [-24.0, +25.0]$ m^2/s^2

Figure 1: Stokes flow verification in the lid-driven benchmark showing viscous dissipation and symmetric pressure extrema ($t = 0.1$ s).

4.2 Method 2: Newtonian Annular Couette Verification



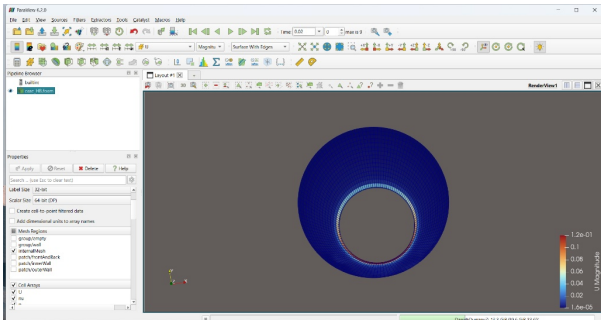
(a) Horizontal Velocity $U_x \in [-0.12, +0.15]$ m/s



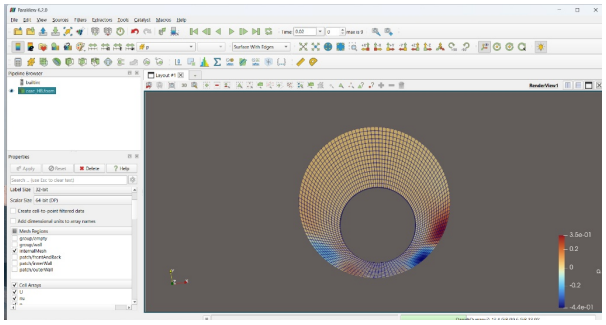
(b) Kinematic Pressure $p \in [-7.8 \times 10^{-4}, +6.4 \times 10^{-3}]$ m^2/s^2

Figure 2: Newtonian eccentric annulus results under 60 RPM rotation demonstrating radial momentum diffusion across the entire annular space ($t = 0.2$ s).

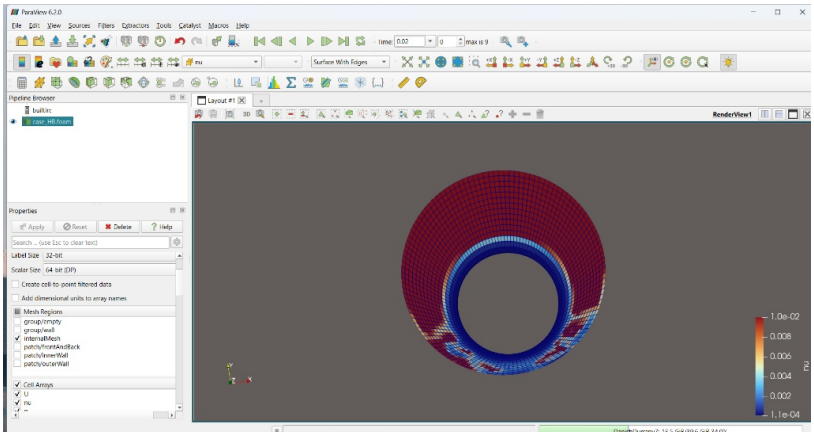
4.3 Method 3: Herschel–Bulkley Drilling Mud Rheology



(a) Velocity Magnitude $|\mathbf{U}| \in [1.6 \times 10^{-5}, 0.12]$ m/s



(b) Kinematic Pressure $p \in [-0.44, +0.35]$ m^2/s^2



(c) Effective Kinematic Viscosity $\nu \in [1.1 \times 10^{-4}, 1.0 \times 10^{-2}]$ m^2/s

Figure 3: Herschel–Bulkley simulation exhibiting sharp shear-thinning in the narrow gap and a static unyielded plug core in the wide upper annulus ($t = 0.2$ s).

5 Physical Insights & Multiphase Cuttings Cleaning Significance

The comparative analysis of the three fluid governing equations provides critical verification benchmarks for drilling hydraulics:

1. **Unyielded Upper Annulus Plug Core:** In the wide upper clearance of the eccentric annulus ($e = 0.623$), shear rates fall below the threshold required to overcome the fluid yield stress ($\tau < \tau_y$). Consequently, the mud maintains its upper viscosity limit ($\nu_0 = 1.0 \times 10^{-2}$ m^2/s). This rigid plug provides vital static cuttings suspension capability, preventing settling during pump shut-offs.

2. **Narrow Clearance Shear Thinning:** Near the rotating drillpipe and within the constricted lower gap, local shear stresses exceed the yield point ($\tau \gg \tau_y$). The effective viscosity drops by two orders of magnitude ($\nu \approx 1.1 \times 10^{-4}$ m^2/s). This localized shear liquefaction significantly reduces rotational torque while generating hydrodynamic lift forces capable of re-entraining settled cuttings beds into the primary axial transport stream.